Fine-tuning DeBERTaV3 with Adaptive Training Strategies for Fine-Grained Sentiment Analysis

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***Abstract*—Artificial intelligence (AI) plays a pivotal role in modern society, especially in business intelligence, where it drives insights for customer feedback analysis, targeted recommendations, and strategic marketing. Within AI, natural language processing (NLP) enables machines to interpret and analyze human language, with sentiment analysis being a vital subset focused on understanding nuanced emotions in text. This research introduces Fine-tuned DeBERTaV3 with Adaptive Training Strategies (FiTDeBERTaV3-ATS), a model designed for fine-grained sentiment analysis that integrates DeBERTaV3 with an attention mechanism, cross-fold training, and multi-sample dropout. Evaluated on the fine-grained Stanford Sentiment Treebank (SST-5) dataset, the proposed model achieved an improved accuracy of 62.40%, surpassing existing baselines and highlighting DeBERTaV3's effectiveness for this task. These tailored strategies address the unique challenges of SST-5, such as figurative language and a limited dataset size, significantly enhancing the model's ability to capture subtle sentiment distinctions.**

***Index Terms*—Fine-grained sentiment analysis, DeBERTaV3, Fine-tuning, Adaptive training strategies.**

I. Introduction

In the digital era, people share insights and experiences related to products, services and events through social networks and online forums. Potential customers rely on this collective wisdom to guide their decisions, but the variability in user-generated reviews makes it difficult to identify broader patterns and trends [1]. Sentiment analysis addresses this challenge by interpreting emotions in text, enabling businesses to gauge customer satisfaction, refine strategies, and make data-driven decisions. It also equips researchers with tools to monitor societal trends and public opinions at scale. The growing interest in this field is evident by the 34% annual increase of the phrase "sentiment analysis in social networks" in publications from 2008 to 2022 [2]. Initially focused on simple binary or coarse-grained classification—identifying sentiments as positive, negative, or neutral—sentiment analysis has evolved into fine-grained approaches that capture subtle emotional variations [3], [4]. Expanding sentiment categories to include negative, somewhat negative, neutral, somewhat positive, and positive allows researchers to uncover deeper insights into public opinions, consumer behaviors, and social media trends [5]. This granularity is crucial for business intelligence applications such as targeted marketing, personalized recommendations, and product development [6]. However, fine-grained sentiment analysis presents unique challenges, including the need for significantly larger datasets, more sophisticated algorithms capable of detecting subtle changes, and strictly calibrated labeling protocols [7]. Developing reliable methods for fine-grained sentiment analysis remains a key focus of current research, bridging the gap between coarse opinion mining and deeper understanding of sentiment nuances in computational systems.

Traditional models like Support Vector Machine (SVM) and Naive Bayes have demonstrated strong performance in binary sentiment classification [8] but have not been extensively studied for fine-grained sentiment analysis. Deep learning models such as Convolutional Neural Networks (CNNs) achieved 90% accuracy on the binary Stanford Sentiment Treebank (SST-2) but dropped to 53.4% on the fine-grained SST-5 dataset [9]. Transformer-based models, such as Bidirectional Encoder Representations from Transformers (BERT) and its enhanced versions like Robustly Optimized BERT Pretraining Approach (RoBERTa), have excelled in NLP tasks by capturing bidirectional contextual meaning [10]. Recently, Decoding-enhanced BERT with Disentangled Attention (DeBERTa) built upon the strengths of BERT and RoBERTa, introducing disentangled attention mechanisms and other key innovations [11]. Its latest iteration, DeBERTaV3, introduces architectural improvements and a more effective pretraining strategy, making it a strong candidate for fine-grained sentiment analysis and other natural language understanding tasks. [12].

This research presents an optimized DeBERTaV3-based classifier for fine-grained sentiment analysis. By integrating key techniques such as attention mechanisms, cross-fold training, and multi-sample dropout, the model effectively captures subtle sentiment nuances. Its strong performance on the SST-5 dataset highlights its potential to advance AI-driven business intelligence and decision-making.

II. Related Works

A. Transformer-Based Models

Transformer-based models have set new standards in NLP by capturing long-term dependencies and enhancing word

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relationships within sentences using multi-head attention and positional embeddings, making them highly effective for tasks like fine-grained sentiment analysis. Fine-tuned $BERT_{LARGE}$ achieves a test accuracy of 55.4% on SST-5, while $RoBERTa_{LARGE}$ improves this to 58.2% without additional fine-tuning [10]. Increased performance is attainable through targeted training strategies or adjustments to the base model architecture.

For example, the $RoBERTa_{LARGE}$ and Self-Explaining model embeds an interpretation layer directly into the model, aggregating information across spans and weighting phrases for predictions [13]. This method improves both performance and interpretability, achieving 59.1% accuracy on SST-5. Similarly, Heinsen Routing with $RoBERTa_{LARGE}$ embeddings refines token prioritization by dynamically adjusting token importance across layers, focusing on sentiment-critical elements. This approach reaches 59.8% accuracy, demonstrating its effectiveness in capturing subtle sentiment distinctions [14].

B. Decoding-enhanced BERT with Disentangled Attention (DeBERTa)

DeBERTa, shown to achieve superior performance in NLP tasks, builds on BERT and RoBERTa by introducing key innovations that improve model performance and contextual understanding [11]. A major improvement is the Disentangled Attention Mechanism, which represents each word using two separate vectors—one for content and another for relative position—unlike BERT's single combined vector (Fig. 1). Disentangled matrices compute attention weights, modeling content-to-content and content-to-position interactions independently. This separation improves the model's ability to handle relative positions, capturing structural nuances in long or complex sequences.

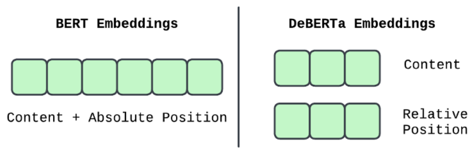

Fig. 1: Processing Embeddings in BERT and DeBERTa

While relative positioning is crucial, absolute positional embeddings are also important for contextual understanding. Unlike BERT, which integrates absolute positional embeddings in the input layer, DeBERTa introduces absolute positioning at the decoding stage, just before the softmax layer, in a design called the Enhanced Mask Decoder (Fig. 2, illutrastion adapted from He et al. (2021) [11]). This approach, which maintains relative positioning throughout the transformer layers, has been shown to outperform methods that embed absolute positions earlier in the model [11]. Lastly, DeBERTa incorporates Scale-Invariant Fine-Tuning (SiFT), an adversarial algorithm that enhances generalization by applying small perturbations to normalized embeddings, improving robustness and stability during fine-tuning.

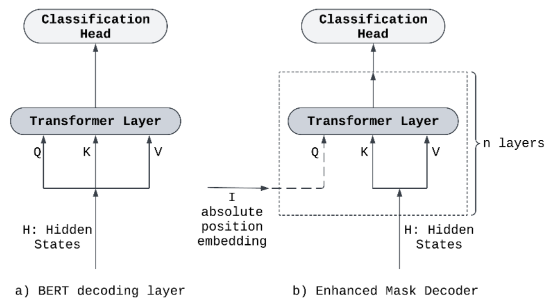

Fig. 2: Decoding in BERT and DeBERTa (Illustration adapted He et al. (2021) [11])

C. DeBERTaV3: Improving DeBERTa

DeBERTaV3 outperformed DeBERTa, achieving a 1.37% higher average score on the General Language Understanding Evaluation (GLUE) dataset—a benchmark for natural language understanding [12]. This improvement is achieved by replacing the Masked Language Modeling (MLM) objective with Replaced Token Detection (RTD) and incorporating Gradient-Disentangled Embeddings Sharing (GDES). Below is a detailed breakdown of these components.

1) Masked Language Modeling (MLM) trains models by masking tokens and predicting them using the surrounding context. While effective for learning contextual embeddings, it suffers from inefficiencies as only masked tokens contribute to the loss, requiring more iterations and leading to slower convergence due to redundant computations for unoptimized tokens.
2) Replaced Token Detection (RTD) replaces MLM as the primary pretraining task. This approach trains a generator to substitute tokens with plausible alternatives and a discriminator to distinguish between original and replaced tokens. Unlike MLM, RTD utilizes all tokens during training, improving data efficiency and accelerating learning. To enhance the quality of token replacements, the generator is also trained with MLM. This dual-objective method improves semantic representation while significantly boosting training efficiency.
3) Gradient-Disentangled Embeddings Sharing (GDES) combines the strengths of two embedding approaches: Embedding Sharing (ES) and No Embedding Sharing (NES). While ES reduces parameters by sharing embeddings between the generator and discriminator, it suffers from gradient conflicts due to opposing objectives—the generator aligns semantically similar tokens for high-quality replacements, while the discriminator separates them for distinction. NES avoids these conflicts by using separate embeddings but at the cost of a significantly increased parameter

size. GDES resolves this trade-off by disentangling gradient updates for shared embeddings, achieving both parameter efficiency and conflict-free training.

DeBERTaV3 offers three model sizes, summarized in Table I. For this experiment, DeBERTaV3$_{\text{LARGE}}$ was selected for fine-tuning and evaluation due to its larger hidden size, increased number of attention heads, and additional layers. These features enhance its ability to capture nuanced sentiment distinctions across the five classes. While this choice requires higher computational resources, the emphasis on achieving superior performance justifies the trade-off.

TABLE I: Configurations of DeBERTaV3 Model Sizes [12]

Model	Hidden Size	Attention Heads	Layers	Parameters
DeBERTaV3$_{\text{SMALL}}$	768	12	6	86M
DeBERTaV3$_{\text{BASE}}$	768	12	12	140M
DeBERTaV3$_{\text{LARGE}}$	1024	16	24	350M

III. Proposed Method

The proposed model, fine-tuned DeBERTaV3 with adapted training strategies (FiTDeBERTaV3-ATS), is designed to address the challenges of fine-grained sentiment analysis. Its training strategies include an attention mechanism, cross-fold training, and multi-sample dropout.

1) Attention Mechanism
To identify sentiment-critical features, the model employs a learned attention mechanism that generates attention weights through a neural network with LayerNorm and GELU activations. The weighted token embeddings are concatenated with the [CLS] token representation to create a combined feature vector, capturing both local and global context.
2) Cross-Fold Training
A k-fold stratified cross-training strategy is employed to maintain reliable label distributions, improve generalization, and reduce error rates [15]. Applied as an ensemble learning technique, it trains multiple model instances on different subsets, enhancing robustness and limiting dependence on any single split. Final predictions are averaged across trained instances for more diverse data exposure and optimized hyperparameters.
3) Multi-Sample Dropout
Dropout prevents overfitting by randomly deactivating neurons during training. Multi-sample dropout extends this process by applying multiple dropout rates simultaneously, introducing feature perturbations that can improve performance of transformer models [16]. Each variation of the transformed feature vector is processed independently by classification heads to generate sentiment logits. Predictions are then aggregated across multiple paths for more reliable results. Combined with cross-fold training, this method has been shown to enhance model results [16].

The high-level architecture is illustrated in Fig. 3. First, data

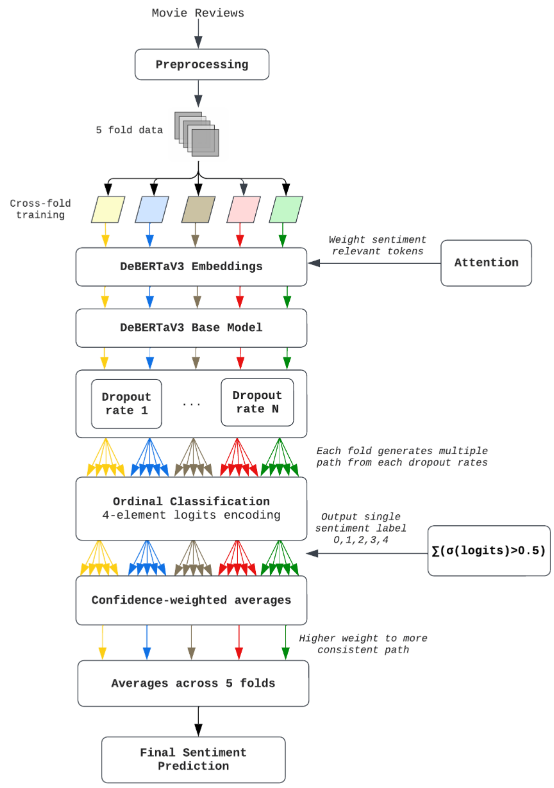

Fig. 3: Architecture of the proposed model - fine-tuned DeBERTaV3 with adapted training strategies (FitDeBERTaV3-ATS)

is split using a 5-fold cross-training strategy. Preprocessing includes removing special characters, punctuation, HTML tags, and normalizing to lowercase. Input sequences are then tokenized using DeBERTaV3 tokenizer and processed through the DeBERTaV3 base model to generate contextual embeddings. These embeddings are further refined through an attention mechanism that integrates attended features with the [CLS] token for classification. To enhance generalization, multi-sample dropout is applied, generating diverse feature representations by varying dropout rates.

In each dropout path, the model employs ordinal classification to preserve the hierarchical structure of sentiment labels (negative $<$ somewhat negative $<$ neutral $<$ somewhat positive $<$ positive). During training, input labels are transformed into binary threshold vectors, where each element represents whether the sentiment surpasses a given threshold (e.g., a neutral label (2) is represented as [1,1,0,0]). The model generates four logits, which are passed through a sigmoid activation to obtain probability scores. Binary cross-entropy loss is then calculated between these probabilities and the binary targets. At inference, the model produces four ordinal logits corresponding to the decision boundaries, with the final sentiment class determined as:

$$\text{sentiment label} = \sum(\sigma(\text{logits}) > 0.5)$$

For example, sigmoid-transformed logits of [0.8, 0.7, 0.3, 0.1] produce binary predictions [1,1,0,0], summing to 2 for neutral sentiment. Predictions across dropout paths are combined using confidence-weighted averaging, with higher weights assigned to predictions showing greater consistency across paths. As a final step, predictions from all five folds are averaged to determine the final sentiment class.

IV. Experiments and Results

A. Datasets

The proposed approach is evaluated on SST-5, a benchmark dataset designed for fine-grained sentiment analysis. This dataset was first introduced by Pang and Lee (2008) [4] and consists of 10,662 movie review excerpts from rottentomatoes.com, with reviews originally labeled as positive or negative. The dataset was later expanded by Socher et al. (2013) to include a total of 11,855 single sentences and 215,154 phrases parsed from the original reviews using Stanford Parser [17]. The parsing process split some snippets into multiple sentences, resulting in additional sub-phrases. These phrases were annotated with fine-grained sentiment labels by human annotators via Amazon Mechanical Turk. Annotators used an interface with a slider of 25 positions to assign sentiment labels ranging from very negative to very positive. It was observed that most annotations fell into five primary positions: negative, somewhat negative, neutral, somewhat positive, and positive. The 25 original positions were consolidated into these categories to better capture the main sentiment variability.

Fig. 4 illustrates the distribution of sentiment labels across the training, validation, and test sets in the SST-5 dataset. The consistent color proportions within each bar show that the label distribution is comparable across the three splits, ensuring the dataset is well-balanced for training and evaluation. Labels 0 and 4, representing extreme negative and positive sentiments respectively, have the smallest proportions. Labels 1 and 3, corresponding to somewhat negative and somewhat positive sentiments, are more frequent, with label 1 having the highest proportion across all splits. Label 2, representing neutral sentiment, also constitutes a significant portion of the dataset.

Table II summarizes key statistics of the SST-5 dataset, providing an overview of its structure across the training, validation, and test splits. Reviews are short, ranging from 1 to 2 sentences on average and an average word count of approximately 19 words per review across all splits and a maximum word count ranging from 49 to 52. The table highlights that the dataset is consistent across all splits, ensures that the model can be effectively trained, validated, and tested on balanced data for a reliable performance evaluation.

Additionally, examples from each sentiment class in the dataset are presented in Table III, highlighting the variety of expressions used to convey sentiment. The dataset includes a mix of straightforward and nuanced language, often incorporating figurative elements like metaphors, analogies, and sarcasm. These subtle sentiment cues often blur the distinctions between classes, making it difficult to interpret and establish clear boundaries. This inherent complexity highlights the challenges of fine-grained sentiment analysis.

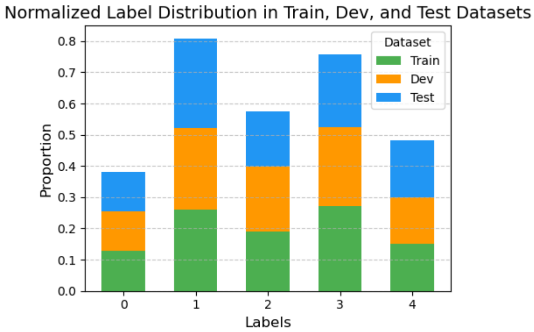

Fig. 4: Label distributions for SST-5 across train, validate, and test set. Labels 0 through 4 represent sentiments ranging from negative (0) to positive (4)

TABLE II: Data Statistics: #s denotes the number of sentences, and #w denotes the number of words.

Split	Size	Avg #s	Max #s	Avg #w	Max #w
Train	8,544	2.19	13	19.14	52
Validation	1,101	2.15	7	19.32	49
Test	2,210	2.17	7	19.19	50

TABLE III: Examples of Sentiment Classes in SST-5

Class	Examples
0 (Negative)	1. A yawn-provoking little farm melodrama. 2. Doesn't add up to much.
1 (Somewhat Negative)	1. A cartoon? 2. The fact that the "best part" of the movie comes from a 60-second homage to one of Demme's good films doesn't bode well for the rest of it.
2 (Neutral)	1. The beautiful, unusual music is this film's chief draw, but its dreaminess may lull you to sleep. 2. The story is so light and sugary that were it a Macy's Thanksgiving Day Parade balloon, extra heavy-duty ropes would be needed to keep it from floating away.
3 (Somewhat Positive)	1. Smart and fun, but far more witty than it is wise. 2. An engaging, formulaic sports drama that carries a charge of genuine excitement.
4 (Positive)	1. An action/thriller of the finest kind, evoking memories of *Day of the Jackal*, *The French Connection*, and *Heat*. 2. A thoughtful what-if for the heart as well as the mind.

B. Model and Training Configurations

The training setup in the proposed model is designed to optimize performance while maintaining stability and enhancing generalization. Input text is preprocessed and then tokenized using the DeBERTaV3 tokenizer, truncating or padding sequences to length of 128 tokens. Multi-sample dropout applies dropout rates from 0.1 to 0.5, with predictions combined using confidence-weighted averaging. Confidence is based on how far the outputs are from 0.5, which represents uncertainty in sigmoid activation. The farther a value is from 0.5, the more confident the model is in its prediction.

Predictions where all four probabilities are consistently high or low across classifiers are considered more reliable. The confidence scores are then scaled using a softmax function with a temperature of 0.5, which adjusts how much influence each prediction has. A lower temperature gives more weight to confident predictions, ensuring the final result is more stable.

The model uses gradient checkpoint to reduce memory usage during training. A cosine learning rate schedule with a 10% warm-up phase gradually increases the learning rate before decaying. Mixed precision training with PyTorch's autocast and GradScaler lowers memory consumption and speeds up GPU computations, while gradient clipping (max norm of 1.0) helps stabilize training. Validation runs at the end of each epoch, calculating loss, accuracy, and per-class accuracy to evaluate performance and identify areas for improvement. Training uses a batch size of 16, with shuffled data loaders and four worker processes for efficient data loading, while validation employs unshuffled data loaders. The model saves the best validation accuracy per fold, and early stopping prevents overfitting by stopping training when validation performance stops improving.

The AdamW optimizer is used, with pin memory enabled in data loaders for faster GPU data transfer. The key hyperparameters are summarized in Table IV. Different learning rates (1×10^{-5}, 2×10^{-5}, 3×10^{-5}) and epoch counts (2, 3, 5) were explored, with 2×10^{-5} and 3 epochs yielding the best performance.

TABLE IV: Hyperparameters Used in Training

Hyperparameter	**Value**
Learning Rate	2×10^{-5}
Max Sequence Length	128
Number of Folds	5
Batch Size	16
Number of Epochs	3
Weight Decay	0.01
Dropout Rates	0.1 to 0.5
Temperature Scaling	0.5

C. Results and Discussion

1) Baseline Comparison on SST-5: The performance of the proposed FiTDeBERTaV3-ATS model is evaluated on the SST-5 dataset. It builds upon DeBERTaV3 for fine-grained sentiment analysis by integrating three key components: an attention mechanism for detecting sentiment-critical features, 5-fold cross-training to improve robustness, and multi-sample dropout with temperature-scaled confidence weighting for improved generalization. Initial experiments evaluated a model incorporating only the attention mechanism. Table V presents the accuracy of the baselines, the initial experiment model, and the proposed model. While the attention-enhanced DeBERTaV3 outperformed existing baselines, including Heinsen Routing + RoBERTa$_{\text{LARGE}}$ (59.8%) and RoBERTa$_{\text{LARGE}}$ + Self-Explaining (59.1%), the performance improvement was modest. This led to the exploration and integration of additional strategies.

TABLE V: Accuracy results on SST-5. The best results obtained from the proposed model are in **bold**. The best published results are underlined

Methods	**Accuracy (%)**
Fine-tuned BERT$_{\text{LARGE}}$ [10]	55.4
RoBERTa$_{\text{LARGE}}$ [10]	58.2
RoBERTa$_{\text{LARGE}}$ + Self-Explaining [13]	59.1
Heinsen Routing + RoBERTa$_{\text{LARGE}}$ [14]	<u>59.8</u>
Attention-enhanced fine-tuned DeBERTaV3	60.27
Proposed Model (FitDeBERTaV3-ATS)	**62.40**

By combining the attention mechanism, cross-fold training, and multi-sample dropout with confidence-weighted averaging and ordinal classification, FitDeBERTaV3-ATS achieved an accuracy of 62.40%. This marks a 2.13% improvement over the initial model that fine-tuned DeBERTaV3 with only the attention mechanism. The integration of these additional strategies significantly enhanced the model's performance. Moreover, FitDeBERTaV3-ATS outperformed other baselines with improvements of 4.2% and 7.0% over RoBERTa$_{\text{LARGE}}$ and BERT$_{\text{LARGE}}$ respectively. These results demonstrate the effectiveness of combining advanced training strategies to better capture fine-grained sentiment distinctions and achieve stronger generalization across different data splits.

Fig. 5 presents the normalized confusion matrix for the performance of FitDeBERTaV3-ATS on SST-5, where rows correspond to true sentiment classes and columns correspond to predicted classes.

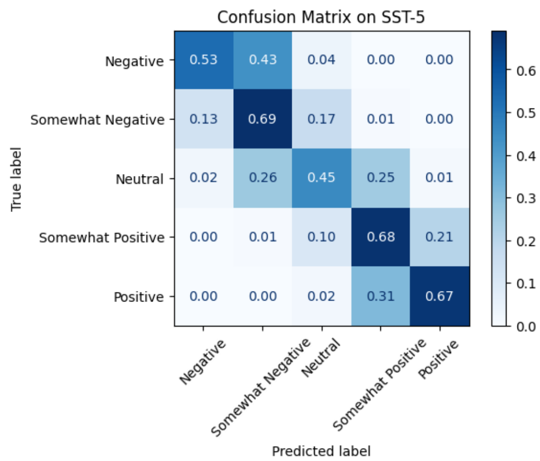

Fig. 5: Confusion Matrix of FitDeBERTaV3-ATS on SST-5

The confusion matrix shows normalized prediction rates per class, with the model performing strongest on somewhat negative (69%), somewhat positive (68%), and positive (67%) sentiments. Neutral sentiment classification (45%) is the weakest, often misclassified as somewhat negative (26%) or somewhat positive (25%), reflecting its ambiguous nature.

The matrix reveals a pattern of confusion between adjacent sentiment levels, notably with 43% of negative samples misclassified as somewhat negative and 21% of somewhat positive samples as positive. These confusions can occur due to subtle linguistic nuances and contextual interpretations that can make sentiment boundaries less distinct. For instance, phrases might contain mixed sentiment signals or require deeper contextual understanding to accurately classify their intensity level. While the overall results demonstrate the model's effectiveness in capturing sentiment gradients and providing valuable insights, they also underscore the inherent complexity of fine-grained distinctions in sentiment analysis, highlighting the need for continued advancements to further enhance precision.

V. Conclusion and Future Work

In this study, the proposed model FitDeBERTaV3-ATS achieved an improved accuracy on the SST-5 dataset, outperforming other baseline models. This demonstrates the effectiveness of DeBERTa over other transformer-based architectures, such as BERT and RoBERTa, for fine-grained sentiment analysis. In addition, it is crucial to utilize advanced training strategies tailored to the specific task and dataset to improve performance. In this research, the SST-5 dataset poses unique challenges, as the use of short but figurative language creates subtle distinctions between sentiment classes. Moreover, the relatively small size of the data set limits the model's ability to generalize effectively without overfitting. The proposed model addresses these challenges through multiple complementary approaches: the attention mechanism helps capture sentiment-critical features in sophisticated language with subtle cues, ordinal classification effectively models the hierarchical nature of sentiment classes, multi-sample dropout improves regularization through diverse feature perturbations, and cross-fold training enhances robustness by exposing the model to varied data distributions.

While the proposed model has achieved significant results, there are still opportunities for further exploration. Future work could include ablation studies to quantify the contribution of each component—attention mechanism, multi-sample dropout, and cross-fold training—to the model's performance. Testing the model's adaptability to different domains and language styles, such as social media posts and informal conversations, would help validate its robustness. Additionally, exploring business applications, such as user satisfaction rankings on platforms like Yahoo Answers as shown by Banjar et al. (2024) [18], could demonstrate its practical value. The continued advancement of fine-grained sentiment analysis remains important for business intelligence applications, enabling more nuanced insights for strategic decision-making in areas like consumer sentiment analysis and public opinion monitoring.

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